reveal a space-filling, intra-cluster gas at tens of millions of degrees. The gas is so hot that it glows strongly in the X-ray part of the spectrum. The very movement of gas-rich galaxies through this medium eventually strips them of their own gas, forcing them to forfeit their capacity to make new stars. That could explain it. But when you calculate the total mass present in this heated gas, for most clusters it exceeds the mass of all galaxies in the cluster by as much as a factor of ten. Worse yet, clusters are overrun by dark matter, which happens to contain up to another factor of ten times the mass of everything else. In other words, if telescopes observed mass rather than light, then our cherished galaxies in clusters would appear as insignificant blips amid a giant spherical blob of gravitational forces.

In the rest of space, outside of clusters, there is a population of galaxies that thrived long ago. As already noted, looking out into the cosmos is analogous to a geologist looking across sedimentary strata, where the history of rock formation is laid out in full view. Cosmic distances are so vast that the travel time for light to reach us can be millions or even billions of years. When the universe was one half its current age, a very blue and very faint species of intermediate-sized galaxy thrived. We see them. They hail from a long time ago, representing galaxies far, far away. Their blue comes from the glow of freshly formed, short-lived, high-mass, high-temperature, high-luminosity stars. The galaxies are faint not only because they are distant but because the population of luminous stars within them was thin. Like the dinosaurs that came and went, leaving birds as their only modern descendant, the faint blue galaxies no longer exist, but presumably have a counterpart in today's universe. Did all their stars burn out? Have they become invisible corpses strewn throughout the universe? Did they evolve into the familiar dwarf galaxies of today? Or were they all eaten by larger galaxies? We do not know, but their place in the timeline of cosmic history is certain.

With all this stuff between the big galaxies, we might expect some of it to obscure our view of what lies beyond. This could be a problem for the most distant objects in the universe, such as quasars. Quasars are super-luminous galaxy cores whose light has typically been traveling for billions of years across space before reaching our telescopes. As extremely distant sources of light, they make ideal guinea pigs for the detection of intervening junk.

Sure enough, when you separate quasar light into its component colors, revealing a spectrum, it's riddled with the absorbing presence of